Miles Jennings

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Education

University of California, Irvine Irvine, CA
B.S. Computer Engineering — Dean's Honor List *Sept. 2022 – June 2026*
Relevant Coursework: Electronics I & II, Network Analysis I & II, Semiconductor Devices, CMOS Circuit Design, VLSI Design, Digital Logic Design, Computer Architecture
Organizations: HyperXite, FUSION, FASAE

Experience

Test Engineer Intern Buena Park, California
Leach International Corp. *March 2026 – Present*
- Diagnosed and repaired **8** malfunctioning test systems within **3** months by analyzing undocumented system architectures and troubleshooting hardware test fixtures, reducing operator downtime and eliminating production bottlenecks across the test floor
- Designed and deployed an automated **Python**-based capacitor validation and conditioning system integrating an **LCR6002**, high-voltage power supply, and safety analyzer, eliminating outsourced testing for **490 components** and reducing validation costs by **60%**
- Authored detailed electrical wiring diagrams in **Altium** to re-engineer and restructure key component testing machinery, simplifying **signal routing** and reducing future system troubleshooting time

Control Systems Engineer Irvine, California
HyperXite – SpaceX Hyperloop *Aug. 2025 – June 2026*
- Evaluated the team's **Raspberry Pi 5**-based architecture against an **STM32H753ZI** solution, selecting the STM32 for lower power consumption and native **CAN/ADC** support, eliminating external interface hardware and improving embedded system reliability
- Tested and integrated hardware components for critical pod safety monitoring, designing custom **breadboard test setups** and interfacing sensors with an **STM32 microcontroller** using **C/C++** and STM32CubeIDE
- Designed a custom **control PCB** using **Altium** to integrate embedded control hardware, power distribution, and sensor interfaces, consolidating multiple subsystems onto a single board to simplify system integration and improve maintainability

Technology Intern Long Beach, California
P2S Engineering *June 2025 – March 2026*
- Designed and coordinated **low-voltage infrastructure systems** for commercial facilities, resolving multidisciplinary design conflicts between electrical, mechanical, and architectural teams to improve constructability before construction handoff
- Cut regulatory revision cycles by **11%** by implementing design fixes flagged by supervisors and principal engineers in **3D models**, improving design accuracy and shortening project timelines

Projects

Zener Diode Voltage Regulator & RC Circuit Analysis | *LTSpice, Circuit Analysis* *Jan. 2025*
- Designed and simulated **Zener diode voltage regulation and waveform-limiting circuits** in **LTSpice**, analyzing diode breakdown behavior and output voltage across varying input amplitudes
- Evaluated the effects of **load resistance** on Zener regulation, analyzing the relationship between V_{out}, V_Z, I_L, and I_Z under different operating conditions
- Analyzed **RC transient behavior** by varying resistance and capacitance values, demonstrating how the **RC time constant** affects capacitor charging/discharging, waveform smoothing, and ripple reduction

4-Bit Full Adder/Subtractor | *Cadence Virtuoso, Spectre, CMOS Logic* *Nov. 2025*
- Designed **transistor-level CMOS** schematics and physical layout for a 4-bit full adder in **Cadence**, optimizing layout while maintaining functional correctness
- Performed **Spectre** simulations and **DRC/LVS verification** to validate functionality, timing behavior, and compliance with fabrication design rules
- Iteratively refined schematic and layout based on verification results to resolve design rule violations and improve manufacturability

Technical Skills

Languages: Python, C/C++, Java, Verilog, RISC-V, MIPS
Software/Tools: LabVIEW, Revit, AutoCAD, KiCad, Bluebeam, Arduino IDE, VSCode, Vivado, LTSpice, Cadence, Linux, Git, GitHub, STM32CubeIDE, MobaXterm
Hardware: Oscilloscope, Power Supply, Multimeter, Circuit Design & Analysis, Microcontrollers, ESP32, Embedded Systems, PCB Design, Arduino, Power Supplies, FPGA, Spectrum Analyzer